MIL-A-60091A(AR)
11 June 1987
SUPERSEDING
MIL-A-60091 (AR)
1 Aug 1970

MILITARY SPECIFICATION

ADHESIVE FOR BONDING DEMOLITON CHARGES TO STRUCTURAL SURFACES

This Specification is approved for use within the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a one-part, ready to use adhesive, suitable for use in bonding demolition charges to dry, wet and underwater surfaces of all types of structural materials, under all weather conditions.

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-A-48078	-Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-T-60394	-Tape, Pressure-Sensitive Adhesive Film Foam Double-Coated (For use with Ammunition)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament,Munitions and Chemical Command, Attn. AMSMC-QA, Picatinny Arsenal, New Jersey 07806-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-141	- Paint, Varnish, Laquer, and Related Materials, Methods of Inspection, Sampling and Testing

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-286	- Propellant, Solid, Sampling, Examination and Testing
MIL-STD-1190	- Minimum Guidelines for Level C Preservation, Packing and Marking
MIL-STD-1218	- ACS Chemicals

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The adhesive shall be a one-part ethyl cyanoacrylate type material compounded with proper stabilizers to meet the requirements of this specification. The manufacturer shall certify that the adhesive complies with this requirement.

3.2 Color. The color of the adhesive shall be clear when tested as specified in 4.5.1.

3.3 Consistency. The consistency of the adhesive shall be such that it can readily be dispensed from its collapsible metal tube container by being squeezed by hand. The adhesive shall be smooth, clear and free of lumps after both ambient and -40°F conditioning, when tested as specified in 4.5.2.

3.4 Set time. The set time shall be 10 seconds maximum when tested as specified in 4.5.3.

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3.5 Viscosity. The visosity of the adhesive shall be as follows when tested as specified in 4.5.4.

TABLE I. Viscosity

Test Temperature	Viscosity (cp)
Ambient (25 ± 2°F)	1950 ± 200
-40° ± 2°F	1750 min.

3.6 Adhesive strength. The adhesive shall meet the requirements of 3.6.1, 3.6.2, and 3.6.3.

3.6.1 Strength at low temperature. The adhesive bond between the tape faced metal surface of the T-bar or equivalent apparatus (see 4.5.5) with a 1/8 pound weight attached, and the steel, cement building block, and plywood test panels shall not fail for a period of 24 hours at -40°F. Vertical and horizontal positions of the bond shall be tested separately. Compliance shall be determined in accordance with 4.5.5.

3.6.2 Strength on wet surfaces. The adhesive bond between the tape faced metal surface of the T-bar or equivalent apparatus (see 4.5.5) with a 1/8 pound weight attached, and the steel, cement building block, and plywood test panels shall not fail for a period of 24 hours. All test surfaces shall be wet. Vertical and horizontal positions of the bond shall be tested separately. Compliance shall be determined in accordance with 4.5.5.

3.6.3 Strength underwater. The adhesive bond between the tape faced metal surface of the T-bar or equivalent apparatus (see 4.5.5) with a 1/8 pound weight attached, and the steel, cement building block, and plywood test panels shall not fail for a period of 24 hours. All tests shall be performed underwater. Vertical and horizontal positions of the bond shall be tested separately. Compliance shall be determined in accordance with 4.5.5.

3.7 Compatability with explosives. When specifically required in the contract or purchase order application shall be made to Government laboratory or other Government approved facility for the determination of compatability of the adhesive with a particular explosive. Testing shall be performed as specified in 4.5.6.

3.8 Storage stability. The adhesive shall meet the requirements of this speification after a storage period of at least 12 months provided that it has been stored in the original sealed collapsible tubes, original shipping containers and at temperatures not exceeding 70°F.

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3.9 Workmanship. The manufacturer shall implement procedures and controls to assure that the process and the product produced are not compromised by foreign materials and contaminants or any other conditions which may degrade the composition. Determination of foreign materials shall be in accordance with 4.5.7.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection (see 4.3)

b. Quality Conformance Inspection (see 4.4)

4.3 First article inspection.

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4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall be five (5) pounds of adhesive. The sample shall be obtained from the first production lot which has been produced by the contractor using the same production processes, procedures and equipment as will be used in fulfilling the contract. All materials shall be obtained from the same sources of supply as will be used in regular production.

4.3.2 Inspection to be performed. The sample will be subjected by the Government to any or all of the examinations or tests specified in this specification (see MIL-A-48078) and Table II.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER / NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Color			3.2	4.5.1
	Consistency			3.3	4.5.2
	Set Time			3.4	4.5.3
	Viscosity			3.5	4.5.4
	Adhesive Strength			3.6	4.5.5
	Compatability			3.7	4.5.6
	Storage Stability			3.8	1/
	Workmanship			3.9	4.5.7

NOTES:
1/ The manufacturer shall certify that the adhesive will meet this requirement.

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4.3.3. Rejection. See MIL-A-48078

4.3.4 Process control. Submission of process controls to be utilized by the manufacture to assure a quality product shall be made to the technical agency for review on all critical defects not subject to direct, discrete measurement and all major defects specifying sampling plans requiring fewer samples than the routine inspection levels of MIL-STD-105.

4.4 Quality conformance inspection. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1 Examination and testing.

4.4.1.1 Sampling for chemical and physical testing. Representative samples shall be taken by random selection of five containers or boxes from each adhesive lot. Equivalent quantities (number of tubes) shall be taken such that the aggregate quantity shall be sufficient for the required tests. Three samples shall be placed into separate containers, immediately closed and identified with a label showing the container from which the samples were taken, lot number, nomenclature, and packing date. Prior to conducting lot acceptance tests, these samples shall be permitted to attain room temperature (70 $\pm$ 5°F). These tubes of adhesive shall be used for chemical and physical tests. If the sample fails to meet any of the requirements specified in section 3, the lot shall be rejected. Each required test shall be performed in duplicate. The classification of defects shall be as given in Table III.

TABLE III. Classification of defects

Test/Examination	Paragraph	Defects Classification
Color	3.2	Major
Consistency	3.3	Major
Set Time	3.4	Major
Viscosity	3.5	Major
Adhesive Strength	3.6	Major
Compatability	3.7	Major

4.5 Methods of inspection.

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4.5.1 Color. The adhesive shall be squeezed from its tube onto a clean surface and visually examined for compliance with the requirement.

4.5.2 Consistency. Two unopened tubes of the adhesive are required for this determination. One tube shall be placed at ambient (70 ± 2°F) conditions for 4 hours. The second tube shall be placed in a low temperature chamber maintained at -40° ± 2°F for 4 hours. Both tubes shall then be removed from the conditioning, allowed to come to room temperature, and the adhesive contents of each squeezed out by hand onto clean, smooth glass surfaces. The adhesive should be spread by means of wooden tongue depressors and visually inspected.

4.5.3 Set time.

4.5.3.1 Apparatus.

a. 2" x 1/2" x 1/16" bright mild steel test plates
b. 5 lCentaur pipette
c. Scotchbrite pad
d. Acetone
e. 1,1,1-Trichloroethane
f. Stopwatch

4.5.3.2 Procedure

a. Degrease the oily test pieces in 1,1,1-Trichloroethane and dry on a paper towel

b. Rub the bonding surfaces lightly with scotchbrite to remove any discoloration. Rinse well with acetone and dry on a paper towel. Rinse for a second time in clean acetone and allow the solvent to evaporate (about 30 sec.) at room temperature. Place the test pieces in the humidity cabinet (50% RH) for 15-30 minutes prior to use.

c. Using a 5μ l Centaur pipette, place 1 drop of adhesive in the center of one of the test pieces and immediately cover with the other to form a T-shaped assembly. Start the stopwatch.

d. Hold the plates firmly with finger pressure then attempt to break the bond manually by pulling in shear.

e. If the bond fails, then repeat the test for further 5 sec increments and determine the actual set time.

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4.5.4 Viscosity. The viscosity of the adhesive shall be determined with a Cannon-Fenske modified Ostwald viscometer, or equivalent, in accordance with manufacturer's instructions. The appropriate calibrated constant based on test temperature shall be used to obtain viscosities in centipose.

4.5.5 Adhesive strength. This test is to be performed in conjunction with the double-sided tape (MIL-T-60394).

4.5.5.1 Apparatus and test panels.

a. Apparatus. The apparatus to be used is shown in Fig. 1. The base of the T-bar has one square inch surface area with 1/8 lb weight (or 60g of brass weights) attached to the T-bar. The tape shall be affixed to the base of the T-bar, then the backing removed when ready for testing. The application of adhesive and bonding to the test panels shall be performed as described for each test condition given below.

b. Test panels. The test panels shall be as specified in Table IV. Each panel shall be prepared for use in adhesive strength determinations by first being weathered. Weathering shall be performed by submerging the test panel for 72 hrs in water, then drying for 24 hrs at 100°F in an oven or other temperature controlled chamber.

TABLE IV. Strength test panels

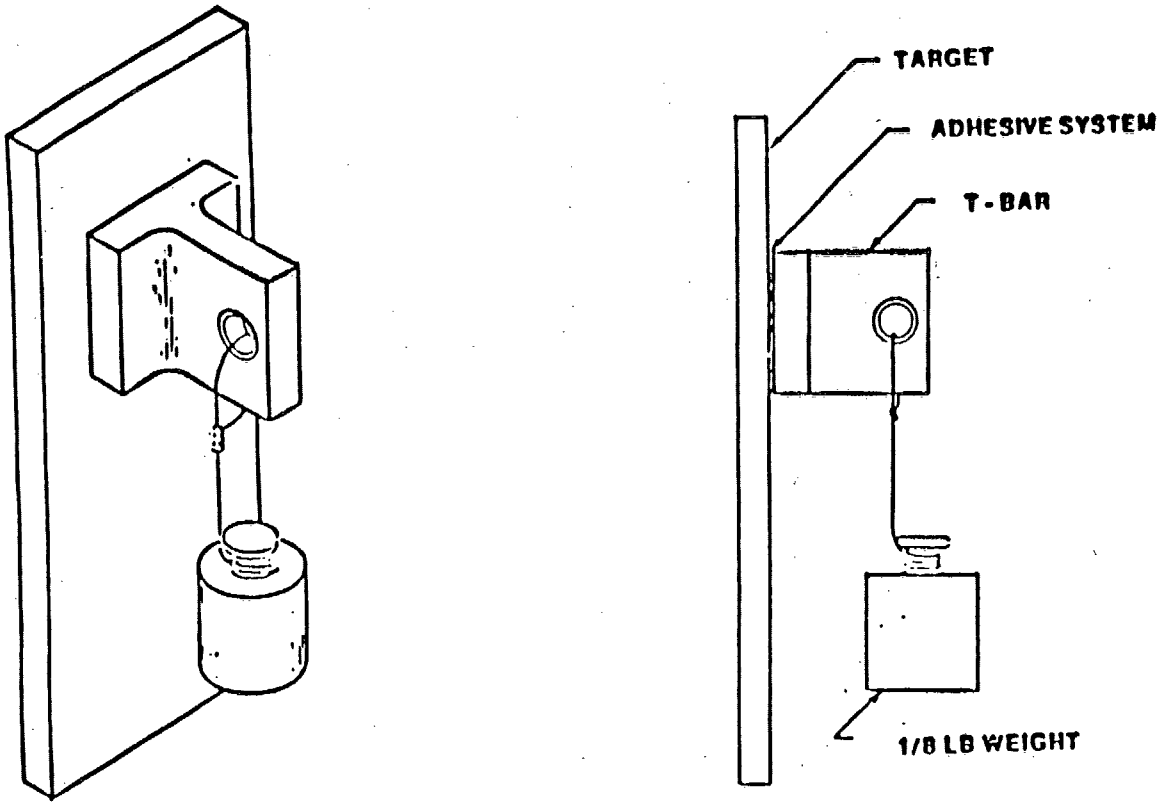
Panel material	Dimensions (inches)			
	Width	Width	Thickness	No. Required
Steel-SAE 1020 cold rolled*1	4	4	.035	40
Plywood, Fir 3-ply exterior grade	4	4	1/4	40
Commercial cement building block*2	4	4	1	40

*1 These panels shall be cleaned in accordance with para 2.1, Method #2011 of FED-STD-141.

*2 The cement block panels shall be cut or chiseled by suitable means from commercial cement building blocks used in the construction trade. Surfaces are to be reasonably uniform.

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T - BAR TEST ASSEMBLY



(Figure 1)

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4.5.5.2 Adhesive strength at low temperature. Each of the test panels (10 of each material) is to be conditioned at -40°F for 4 hours. Any frost that may be present on the panels in the bond area shall be wiped away immediately prior to bonding with the T-bar assembly. The adhesive shall be applied to the tape after the tape backing on the T-bar assembly has been removed. The tape/adhesive surfaces is to be bonded to the panel area by pressing together firmly for approximately 20 seconds. For each of the panel materials, 5 of the panels are to be bonded in the horizontal position (panels parallel to test chamber floor), and 5 in the vertical position (panels perpendicular to test chamber floor). Failure of any adhesive bond to adhere for 24 hours at -40°F shall be cause for rejection of the adhesive lot.

4.5.5.3 Adhesive strength on wet surfaces. Each of the test panels (10 of each material) is to be conditioned by presoaking in tap water for 2 hours. The adhesive is then applied to the tape (without backing), and without wiping, pressed to the test panel as in 4.5.5.2. Vertical and horizontal positions shall be set up as in 4.5.5.2, except that ambient temperatures (70 + 2°F) shall be maintained for 24 hrs. Failure of any adhesive bond to adhere for 24 hrs at this temperature shall be cause for rejection of the adhesive lot.

4.5.5.4 Adhesive strength underwater. Each of the test panels (10 of each material) is to be conditioned by presoaking in clean water having a sodium chloride content of 2.5 percent for 2 hours. The adhesive is then applied to the tape, without backing, while the assemblies are immersed. There should be a head of water of at least 2 inches. Vertical and horizontal positions shall be set up as in 4.5.5.2, except that the temperature shall be maintained at 70° ± 2°F for 24 hrs. The bonds are to be established underwater. Failure of any adhesive to adhere for 24 hrs underwater shall be cause for rejection of the adhesive lot.

4.5.6 Compatability with explosives. Compatability shall be determined in accordance with MIL-STD-286. Method 408.1.

4.5.7 Workmanship. Visually examine the sample for contamination or foreign matter.

5. PACKAGING

5.1 Packaging.

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5.1.1 Level A. Not applicable.

5.1.2 Level C. Unless otherwise specified, the adhesive shall be packaged in 3 gram aluminum tubes. All containers shall be packaged in individual or multiple folding or setup cartons. Each carton shall be closed and sealed to prevent accidental opening.

5.2 Packing. Unless otherwise specified in the contract or purchase order, packing shall be in accordance with MIL-STD-1190. The intermediate pack quantity shall not exceed 150 cartons.

5.3 Marking.

5.3.1 Marking of adhesive. Each tube of adhesive shall bear a label warning against contact with skin tissue, and in case of body contact, to flush with water. Shelf life and all other required markings shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The item covered by this specification is intended for use in bonding demolition charges to structures.

6.2 Ordering data. See MIL-A-48078.

6.3 Contract data requirements. In addition to the normal distribution of records, when the adhesive is procured by the Department of the Army, one copy of all test data shall be forwarded to Commander, US Army Armament Research Development and Engineering Center (ARDEC), ATTN: AMSMC-QAR-R(D), and one copy to ATTN: SMCAR-AEE-BP, Picatinny Arsenal, NJ 07806-5000.

6.4 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: ARDEC, ATTN: AMSMC-QAR-R(D), Picatinny Arsenal, NJ 07806-5000. This description should include, but not be limited to, the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

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6.5 Subject term (key word) listing.

Adhesive
Supplemental adhesive
Demolition charges

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1375-A378)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-60091A (AR)	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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